



MÁSTER EN FINANZAS CUANTITATIVAS



GUÍA DIDÁCTICA Módulo 8

CURSO 2025/26

Módulo 8: Gestión de Carteras de Inversión (Tema 24)

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PRESENTACION

El Máster en Finanzas Cuantitativa se integra por 10 módulos que se agrupan en los siguientes tres cursos de experto.

Curso de Experto en Métodos Computacionales en Finanzas

Módulo 1: Programación en Python y en R
Módulo 2: Modelos de Simulación
Módulo 3: Datos Financieros y Aprendizaje Automático

Curso de Experto en Métodos Econométricos Aplicados en Finanzas

Módulo 4: Modelos de Series Temporales
Módulo 5: Modelos de Volatilidad y Correlación
Módulo 6: Modelos de No linealidad

Curso de Experto en Análisis Financiero y Gestión de Riesgos

Módulo 7: Valoración de Instrumentos Financieros
Módulo 8: Gestión de Carteras de Inversión
Módulo 9: Gestión de Riesgos Financieros
Módulo 10: Teoría del Valor Extremo

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Quantitative Investment

Módulo 8

Gestión de Carteras de Inversión

Active Portfolio Management

Chapter 24

Introduction

- Today we will discuss active portfolio management.
- In particular, we will discuss:
 - Bottom-Up: Treynor-Black Approach;
 - Bayesian Statistics;
 - Top-Down: Black-Litterman Approach;
 - Nonparametric: Almgren-Chriss Approach;
 - Risk Parity Portfolios;
 - Practical Issues; and,
 - Valuing Active Management.

Why Active Management?

- Markowitz-Roy Modern Portfolio Theory (MPT) has a flaw:
- Jensen's Inequality! Suggests we can beat MPT:

$$\underbrace{\max_w E(U(w|X))}_{\text{stochastic optimization}} \geq \underbrace{\max_w U(w|E(X), \text{Var}(X))}_{\text{modern portfolio theory}}. \quad (1)$$

- Example: if $X \sim N(0, 1)$, $[E(X)]^2 = 0 < E(X^2) = 1$.
 - Brinson *et al* suggest we focus on asset allocation.
 - Grinold-Kahn: hold market + asset allocation/active management.
 - Some suggest *smart beta*: hold market, add risk factors.
 - Stochastic optimization is hard; many approximations proposed.
 - Popular: Treynor-Black, Black-Litterman, Almgren-Chriss, risk parity.
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The Treynor-Black Model

- If MPT, single index model are OK: just accommodate alpha.
- That gives us the (bottom-up) *Treynor-Black (1973) model*:
 - 1 Decompose portfolio into market M , active A pieces:

$$r_i = r_f + \alpha_A + \beta_A(R_M + \epsilon_M) + \epsilon_i \quad r_i \perp r_j, R_M \forall i, j. \quad (2)$$

- 2 Put weights w_M , $w_A = 1 - w_M$ on market, active pieces.
- 3 Instrument i active weight $w_i^A = \frac{\alpha_i / \sigma_{\epsilon_i}^2}{\sum_{j=1}^n \alpha_j / \sigma_{\epsilon_j}^2}$.
- 4 Active A , market-neutral alpha A^* portfolio metrics:

$$\alpha_{A^*} = \alpha_A = \sum_{i=1}^n w_i^A \alpha_i, \quad \beta_A = \sum_{i=1}^n w_i^A \beta_i, \quad \sigma_{A^*}^2 = \sum_{i=1}^n (w_i^A)^2 \sigma_{\epsilon_i}^2. \quad (3)$$

- 5 Regroup M , A into total market M^* , market-neutral alpha A^* .

$$M^* = w_M M + \underbrace{w_A \beta_A A}_{\text{market exposure of } A}, \quad A^* = w_A A (1 - \beta_A). \quad (4)$$

Treynor-Black: Weights, Metrics

- Portfolio A^* weight $w_{A^*} = \frac{\alpha_{A^*}/\sigma_{A^*}^2}{E(R_M)/\sigma_M^2 + (1-\beta_A)\alpha_{A^*}/\sigma_{A^*}^2}$.
- Other weights: $w_{M^*} = 1 - w_{A^*}$, and $w_i^{A^*} = w_{A^*} w_i^A$.
- Risky portfolio metrics: $\beta_P = w_{M^*} + w_{A^*}\beta_A$,
 $E(R_P) = \beta_P E(R_M) + w_{A^*}\alpha_A$, and $\sigma_P^2 = \beta_P^2 \sigma_M^2 + w_{A^*}^2 \sigma_{A^*}^2$.
- Alpha portfolio information ratio: $IR_{A^*} = \sqrt{\sum_{i=1}^n \frac{\alpha_i^2}{\sigma_{\epsilon_i}^2}}$.
- Portfolio Sharpe ratio: $S_P = \sqrt{S_M^2 + IR_{A^*}^2}$.
- Thus alpha increases our returns, often at lower risk.

Treynor-Black: Issues

The Treynor-Black model can have some problems:

- Active and market portfolios might be large.
- Can and do we want to hold, say, $+5A - 4M$?
- Can generate high α_P , but also high σ_P .
- Effect: leveraged bet on alpha vs market; are we so certain of alpha?

Mean-variance optimization . . . is extremely sensitive to . . . assumptions the investor must provide.
— Black and Litterman (1992).

- Can add penalties on w_A, w_M ; try shrinking alphas.

$$\text{Maximize } S_P - \kappa_L \max(w_A, w_M, 0)^2 - \kappa_S \min(w_A, w_M, 0)^2 \quad (5)$$

where κ_L, κ_S are capital usage penalties.

- Can also try squeezing alphas.

Bayesian Statistics

- We keep hitting problems due to input uncertainty.
 - Worse: We are not even certain how uncertain we are.
 - Also, we have no way to merge in beliefs or biases.
- One solution: *Bayesian statistics*.
 - Uses a *prior distribution* to express *a priori* beliefs.
 - Mix prior with data likelihood to get *posterior distribution*.
 - Loose (“flat” / “weak”) priors let data control posterior.
 - Tighter priors let preconceived beliefs affect posterior.

Bayesian Statistics: Example

- An example: What is $P(\text{coin flip} = \text{heads})$?
- Recall: n flips, $P(\text{head}) = p$, # heads $k \sim \text{Binomial}(n, p)$.
 - *Conjugate prior* (make math easier) for Binomial is Beta.
- Flat prior: $\text{Beta}(\alpha, \beta = 1) = \text{unif}(0,1)$.
- Flip coin 50 times, get 20 heads. Unusual?
- Posterior: $\text{Beta}(k + 1, n - k + 1) = \text{Beta}(21, 31)$.
 - Posterior $E(k) = \frac{\alpha}{\alpha + \beta} = \frac{21}{52} = 0.404$.
 - Posterior $sd(k) = \sqrt{\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}} = \sqrt{\frac{21 \cdot 31}{52^2 \cdot 53}} = 0.07$.
- 95% *credible interval*: $0.5 \in [0.276, 0.539]$. Seems fair.
- Strong prior of fair coin: $\text{Beta}(\alpha = \beta > 1)$.

The Black-Litterman Model

- Treynor-Black hints at difficulty: we want good inputs.
- Black and Litterman (1992): top-down, Bayesian approach.
- Allows merging historical data, equilibrium model, views;
- Can accommodate non-normality of returns, other risk measures;
- Use macro, cyclical factors for return forecasts; and,
- Use higher-frequency data to forecast covariance matrix.
- Bayesian approach squeezes parameters toward mean; sensible.
- Bayesian approach also enables easy stochastic optimization.

Black-Litterman Process

- The Black-Litterman model can be broken into 5 steps:
 - 1 Estimate covariance matrix Σ with recent data.
 - 2 Determine baseline (prior) forecast and precision.
 - 3 Express views quantitatively.
 - 4 Add views to get revised (posterior) forecast and precision.
 - 5 Optimize portfolio using the posterior distribution.
- Typically done with normal dist'n + mean, variance.
- Also usually stocks and bonds; ignore commodities, RE, FX.
- Fatter-tailed dist'ns + coherent risk could be used.

Black-Litterman: Covariance, Baseline (Prior)

1. Estimate Σ : can use recent, frequent data; factor models.
 - Yields $\hat{\Sigma}$ for asset classes: (σ_B^2, σ_S^2) on diagonal, Σ_{BS} off diagonal.
2. Assume market is in equilibrium (efficient) to start.
 - Outstanding bond, stock amounts $\implies$ weights: $w^T = (w_B, w_S)$.
 - Prior portfolio variance: $\sigma_M^2 = w^T \Sigma w = w_B^2 \sigma_B^2 + w_S^2 \sigma_S^2 + 2w_B w_S \Sigma_{BS}$.
 - Mean risk aversion $\bar{\lambda} \stackrel{?}{=} 3 \implies$ asset class risk premia.
 - In equilibrium, risk premia = forecast: $E(R) = \bar{\lambda} \hat{\Sigma} w$.
 - Get forecast covariance $\text{Var}(\hat{R}) = \hat{\Sigma}/n$; $n = \#$ obs. to estimate Σ .
 - Thus our baseline prior: $\hat{R}^{\text{prior}} \sim N(\bar{\lambda} \hat{\Sigma} w, \hat{\Sigma}/n)$.

Black-Litterman: Expressing Views

3. Next, express quantified views Q for picks P .

- e.g. Bonds will outperform stocks by 1%; bond excess returns=3%:

$$\underbrace{\begin{bmatrix} 0.01 \\ 0.03 \end{bmatrix}}_{E(Q)} = \underbrace{\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}}_P \underbrace{\begin{bmatrix} \hat{R}_B \\ \hat{R}_S \end{bmatrix}}_{\hat{R}}. \quad (6)$$

- The shakiest part: guess view uncertainty $\text{Var}(Q) = \Omega$.
- Concern: inept/dishonest/overconfident analyst can bias portfolio.
- Forecast model/empirical Bayes might give better Ω .
- This give us our “data” (views) likelihood: $P\hat{R} \sim N(Q, \Omega)$.

Black-Litterman: Finding the Posterior, Optimizing

4. Use priors, views to get posterior forecast distribution.

- $E(\hat{R}^{\text{post}})$ is precision-weighted mean of baseline, view:

$$\hat{R}^{\text{post}} \sim N\left(\hat{\Sigma}^{\text{post}} \left[n\hat{\Sigma}^{-1}\bar{\lambda}\hat{\Sigma}w + P^T\Omega^{-1}Q \right], \hat{\Sigma}^{\text{post}}\right) \quad (7)$$

$$\hat{\Sigma}^{\text{post}} = \left[n\hat{\Sigma}^{-1} + P^T\Omega^{-1}P \right]^{-1}. \quad (8)$$

- For posterior of returns, add forecast and return variance:

$$R^{\text{post}} \sim N\left(\hat{\Sigma}^{\text{post}} \left[n\hat{\Sigma}^{-1}\bar{\lambda}\hat{\Sigma}w + P^T\Omega^{-1}Q \right], \hat{\Sigma} + \hat{\Sigma}^{\text{post}}\right). \quad (9)$$

5. Then optimize portfolio to find new $w^{\text{post}} = (w_B^{\text{post}}, w_S^{\text{post}})$.

- Could do Markowitz-Roy, but why? Suboptimal; wastes hard work.
- Could also use non-normal distributions, coherent risk.
- Have distributions; do stochastic optimization! Simulate/quadrature.

Almgren-Chriss Approach

- So far, approaches have entailed distributional assumptions.
- Instead, use distribution-agnostic *nonparametric statistics*.
 - Use *rank statistics* $\rho = (\rho_1, \dots, \rho_k)$ wh/order returns $r = (r_1, \dots, r_k)$.
 - i.e. $r_i \geq r_j \iff \rho_i \geq \rho_j \forall i \neq j$.
 - Consider set of all rank-equivalent returns $Q_r = \{r' : \rho(r') = \rho(r)\}$.
- Suppose we compare two portfolios A and B w/weights w^A, w^B .
- Weakly prefer portfolio A vs B ($A \succeq B$ or $w^A \succeq w^B$) if:
 $w^A \rho(r') \geq w^B \rho(r') \forall r' \in Q_r$.
- Can then impose budget constraints: $\sum_i w_i^A = 1, w_i^A \geq 0$, etc.
- Efficient portfolio is P : $P \succeq P'$; P, P' meet all constraints.
- Can find P by using centroids of returns.

Risk Parity Portfolios

- Budget, industry/sector, risk constraints are common.
- *Risk parity*: each instrument/asset contributes equal risk.
- Like a risk-weighted analog of $1/n$ equal-weight portfolio.
- Has been successfully used at PanAgora, Bridgewater.
- For volatility, risk parity vol between min-variance and $1/n$ portfolio.
 - Thus risk parity portfolio is like shrinkage of $1/n$ portfolio.
- Is this a good idea? Maybe low-risk instruments return more?
 - Asness, Frazzini, and Pedersen (2014): low-beta stocks outperform.
 - Pearson; Boudt+Carl+Peterson: outperforms for coherent risk.
 - Outperformance seems to be about risk, not risk measure.
- Problem: over- (under-) weight if risk under- (over-)estimated.
 - Like problem with cap weighting; could squeeze risk measures.

Practical Issues

- A few issues we should also consider/try to fix.
- Estimated vol: need Jensen's Inequality correction.
- Bond portfolios: less liquid, hard to index. Buys often:
 - *substitute* similar higher-yield bond;
 - earn sector/country *intermarket spread*;
 - *anticipate* future interest rates; or,
 - *pickup yield* of illiquidity premium.
- Sell OOM options to collect premium, rebalance portfolio.
- Bayesian adjustment to alphas? (Details follow)
- Model realized alphas on predicted alphas.

Adjusting Alphas

- Can use a Bayesian approach to squeeze alphas.
- Prior: $\alpha_A \sim N(0, \sigma_{\epsilon_A}^2)$ using historical $\sigma_{\epsilon_A}^2 = 0.02$; empirical Bayes!
- Suppose forecast yields $E(\alpha_A) = 0.01, \text{Var}(\alpha_A) = 0.001$.
- Combine prior and forecast (data) to get posterior:

$$\alpha_A \sim N \left(\frac{\frac{0}{\sigma_{\epsilon_A}^2} + \frac{0.01}{0.001}}{\frac{1}{\sigma_{\epsilon_A}^2} + \frac{1}{0.001}}, \frac{1}{\frac{1}{\sigma_{\epsilon_A}^2} + \frac{1}{0.001}} \right) = N(0.0095, 0.00095). \quad (10)$$

- Result is weighted average; squeezes forecast toward prior.
 - Weight by $1/\text{Var}(\text{forecast})$; lower variance = more weight.
- Could later combine results from another analysis.

Performance Evaluation

- If we actively managed a portfolio, we might ask:
 - How successful was that management?
 - How did individual strategies/decisions perform?
 - Could management have been better?
 - Measuring performance is hard; easy to game.
 - Gaming-resistant metrics may be used for *external* managers.
 - Other measures may only be useful for assessing *internal* managers.
 - Key to performance evaluation is disentangling:
 - Risk: Did we take risk and just get lucky?
 - Noise: How much performance is unpredictable ($\downarrow 0$)?
 - Skill: What performance seems reproducible?
-

Looking at Rates of Return

- Actual investors need to consider *dollar-weighted averages*.
 - Especially proper since capital invested may change.
 - DCF-based measure: *internal rate of return*, $IRR :=$

$$\left\{ r : \overbrace{i_0 + \frac{i_1}{1+r} + \frac{i_2}{(1+r)^2} + \dots}^{\text{PV(cashflows in)}} = \overbrace{\frac{o_1}{1+r} + \frac{o_2}{(1+r)^2} + \dots}^{\text{PV(cashflows out)}} \right\}. \quad (11)$$

- Can also scale historical returns for risk taken (as in § 8.4):
 - *Sharpe ratio*: $S_P = \frac{\bar{r}_P - \bar{r}_f}{\sigma_P}$ adjusts for volatility.
 - *Sortino ratio*: $So_P = \frac{\bar{r}_P - \bar{r}_f}{\theta_P}$ adjusts for semideviation (better).
 - *Cond'l Sharpe*: $CS_P = \frac{\bar{r}_P - \bar{r}_f}{ES_P}$ adjusts for expected shortfall (coherent).
 - *Treynor ratio*: $T_P = \frac{\bar{r}_P - \bar{r}_f}{\beta_P}$ adjusts for systematic-relative risk (sort of).
- Returns should always include transactions costs.

Benchmark-Relative Performance Metrics

- More sensible is to compare performance to benchmark M .
- Modigliani-Modigliani: scale volatility to σ_M w/risk-free F .

$$M_P^2 = r_{P^*} - r_M, \text{ where } P^* = \frac{\sigma_M}{\sigma_P} P + \left(1 - \frac{\sigma_M}{\sigma_P}\right) F. \quad (12)$$

- Can look at *tracking error* vs benchmark we track T :
 - Often, people say “tracking error” but mean *target risk* $sd(R_E)$.

$$R_E = R_P - R_T \quad \text{or} \quad sd(R_E) = sd(R_P - R_T). \quad (13)$$

- Better: Factor exposure is cheap to attain; don't pay fees on that.
 - *Jensen's alpha*: $\alpha_P = \bar{r}_P - (\bar{r}_f + \beta_P(\bar{r}_M - \bar{r}_f))$ CAPM-adjusted.
 - Connor-Korajczak: look at α_P from a multi-factor model.
 - *Info ratio*: $IR_P = \frac{\alpha_P}{\sigma_{\epsilon_P}}$ for α_P from factor model; like idiosyncratic S_P .
 - IR_P in 0.4–0.6 range is very good; above 1 is rare for long.

Summary on Performance Metrics

- T_P good for portfolio additions to LARGE diversified portfolios.
 - S_P popular but not great; CAPM is insufficient alpha model.
 - M^2 very good for comparing strategies of varying risk.
 - IR_P good for judging active management.
 - Note: S_P , IR_P are like t -stats; yet values > 2 are rare.
 - Never good: R^2 . Can drive $R^2 \uparrow 1$: add garbage to model.
 - In general, statistical analysis of manager alpha is tough.
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Metrics for Alternative Investments

- How to evaluate *alternatives*: Hedge funds, CTAs, PE/VC?
- Use Hasanhodzic-Lo (6-factor), Fung-Hsieh (7-/8-factor) models?
 - Problem: many of these factors are expensive to trade
- Can also consider simple Jurek and Stafford (2015):
 - Uses Bondarenko (2014) observation that puts are expensive.
 - Puts are insurance; so, consider a put-writing factor.

$$R_i = \alpha_i + \beta_{iM}R_M + \beta_{i,PW}PW + \epsilon_i. \quad (14)$$

- May also include lags of factors.
- Then filter for funds with good/robust α , IR .
- Many caveats with returns and metrics for alternatives:
 - Funds may report CAPM α and IR s but not clarify that.
 - Observability, Peso problems, survivorship, reporting bias.

Market Timing

- How to detect market timing ability?
- Add nonlinear term to model (Treyner-Mazuy, Henriksson-Merton):
 - Significant $\hat{\gamma} > 0$ indicates timing ability.

$$R_P = \alpha + \beta_P(r_M - r_f) + \gamma(r_M - r_f)^2 + \epsilon, \quad (\text{T-M}) \quad (15)$$

$$R_P = \alpha + \beta_P(r_M - r_f) + \gamma(r_M - r_f)^+ + \epsilon. \quad (\text{H-M}) \quad (16)$$

- Another hint: semideviation θ_P much less than volatility σ_P .
- What is value of market timing?
 - Similar to a call (or call and put) option.
 - *i.e.* Get only upside performance — but pay for that.

Evidence from Investment Professionals

- Based on the data, do some investors seem to have skill?
- Analysts ratings are not informative; *changes* are.
 - Analysts who forecast alpha add some value ($M^2 = 2.1\%$)
- Mutual funds are more varied in performance:
 - Small group of funds+bond funds are consistently bad.
 - Most funds are just plays on momentum, have $\alpha < 0$.
 - No evidence of market timing ability either.
 - Could be some skilled managers + return-chasing investors.
- Hedge funds and mutual funds managed side-by-side:
 - Those MFs have alpha, not taken advantage of by HF; *however*,
 - HF managers who add a MF: MF has typical performance; and,
 - MF managers who add a HF: HF has subpar performance.

Time-Varying Portfolio Metrics

- If returns, risk vary across time, metrics may change.
- Changing means and variances add to risk.
 - Like adding *between-group* variation to *within-group* variation.

$$P_{t_0 \rightarrow t_1} : S_P = \frac{0.2}{0.3}; \quad (17)$$

$$P_{t_1 \rightarrow t_2} : S_P = \frac{0.1}{0.15}; \quad (18)$$

$$P_{t_0 \rightarrow t_2} : S_P = \frac{(0.1 + 0.2)/2}{\sqrt{(0.3^2 + 0.15^2)/2}} = \frac{0.15}{0.24} < \frac{2}{3}. \quad (19)$$

- Successful market timers would adjust positions.
 - Get better performance when market goes up and down.

Manipulation of Measures

- Ingersoll *et al* (2007): Measures can be manipulated.
 - Static: write OOM options, put premium in risk-free bonds;
 - Dynamic: add leverage after extremes to mess w/stationarity;
 - Dynamic: add measurement error by smoothing, illiquid holdings.
- They propose four properties of manipulation-proof measure:
 - ① *Functionhood*: yields one score/investment;
 - ② *Scale invariance*: same score for any notional;
 - ③ *Unbiasedness*: only informed investor can improve score (via arb);
 - ④ *Economic consistency*: Must agree w/market equilibrium.
- Thus their (certainty-equivalent) manipulation-proof measure $\hat{\Theta}$:

$$\hat{\Theta} = \frac{1}{(1 - \rho)\Delta t} \log \left(\frac{1}{T} \sum_{i=1}^T \left[\frac{1 + r_t}{1 + r_{f,t}} \right]^{1-\rho} \right), \quad (20)$$

where ρ is the relative risk aversion (usu. near 3).

What Fee is Fair?

- What is a fair fee for good active management?
- Kane, Marcus, and Trippi use Treynor-Black perspective:

$$\% \text{fee} = f_{\text{one-time}} = \frac{S_P^2 - S_M^2}{2\bar{\lambda}} = \frac{IR_A^2}{2\bar{\lambda}} = \frac{\sum_{i=1}^n IR_i^2}{2\bar{\lambda}}. \quad (21)$$

- If $S_P = 1$, $S_M = 0.8$, and $\bar{\lambda} = 3$, $f_{\text{one-time}} = \frac{0.36}{6} = 6\%$.
- Berk: Manager consumes all expected alpha.
- Glode: Manager may charge *more* than alpha!
 - Why? If alpha is supplied in hard times, that has more value.

What Fee is Fair? Mean-Variance Fee

- What is a fair fee for good active management?
- What if we consider fee vs mean-variance utility?
- Investor pays fee f = utility of receiving $\alpha - f$ with volatility $\hat{\sigma}_\alpha$:

$$f = \alpha - f - \frac{\bar{\lambda}}{2} \hat{\sigma}_\alpha^2. \quad (22)$$

- Implies manager claims annual fee as fraction of α :

$$\frac{f_{\text{annual}}^*}{\alpha} = \frac{1}{2} - \frac{\bar{\lambda}\alpha}{4IR_A^2} \quad (23)$$

- Foster and Young: No manipulation-proof fee structure.
- However, we should consider risk, risk aversion.

Performance Attribution

- Evaluating management performance is tough:
 - Need many observations of portfolio returns; and,
 - Parameters (μ, σ) change constantly over time.
- However, some *decompositions* are informative.
 - Break performance into meaningful pieces.
- Measure baseline performance with benchmark
 - Benchmark allocation $\{w_{B_k}\}_{k=1}^K$ among K asset classes.
 - Benchmark asset class weights, returns become *counterfactuals*.
- Benchmark B , portfolio P performance given by:

$$r_B = \sum_{k=1}^K w_{B_k} r_{B_k} \quad r_P = \sum_{k=1}^K w_{P_k} r_{P_k}. \quad (24)$$

Performance Attribution: Decomposition

- Brinson-Hood-Beebower: decompose benchmark-excess returns.
- Idea: benchmark asset-class weights, returns = counterfactuals

$$r_P - r_B = \sum_{k=1}^K \left(\underbrace{w_{P_k} r_{P_k} - w_{P_k} r_{B_k}}_{\text{security/sector selection}} + \underbrace{w_{P_k} r_{B_k} - w_{B_k} r_{B_k}}_{\text{asset class selection}} \right). \quad (25)$$

- Further: Use sector i sub-portfolios $P_{k,i}$, $B_{k,i}$ for asset class k .
- Decompose asset class k sector/security selection $r_{P_k} - r_{B_k}$:

$$r_{P_k} - r_{B_k} = \overbrace{\sum_{s=1}^S \left(\frac{w_{P_{k,s}}}{w_{P_k}} - \frac{w_{B_{k,s}}}{w_{B_k}} \right) r_{P_{k,s}}}^{\text{sector selection}} + \overbrace{\sum_{s=1}^S \frac{w_{B_{k,s}}}{w_{B_k}} (r_{P_{k,s}} - r_{B_{k,s}})}^{\text{instrument selection}}. \quad (26)$$